AISHA KHAN

Senior Applied AI Engineer

Lahore, Pakistan | aisha.khan@example.com | Professional experience: 5 years 9 months

PROFESSIONAL PROFILE

Applied AI engineer with 5 years 9 months of professional experience shipping Python, NLP, deep-learning, and LLM applications. Owns model evaluation from dataset design to production monitoring. Regularly translates research prototypes into observable APIs and products for internal users.

TECHNICAL SKILLS

- Python; scikit-learn; PyTorch; TensorFlow; XGBoost; pandas; NumPy
- Transformers; natural-language processing; entity extraction; text classification; embeddings; RAG
- OpenRouter/OpenAI model APIs; LangChain; LangGraph; tool calling; structured-output validation
- Model evaluation: macro F1, AUROC, recall@k, grounded-answer audits, regression suites
- FastAPI; Docker; Kubernetes; PostgreSQL; Redis; CI/CD; model serving
- Google Cloud Run; Vertex AI; BigQuery; monitoring; Node.js; Firebase

PROFESSIONAL EXPERIENCE

Lead Applied AI Engineer | Meridian Analytics (fictional) | **Jan 2024 - Present**

- Led a six-person team delivering a multilingual support copilot integrated with hosted LLM APIs, LangChain retrieval chains, and LangGraph approval workflows.
- Built Python/FastAPI orchestration services for document ingestion, semantic search, structured responses, and escalation to human reviewers.
- Evaluated answer grounding using 600 manually reviewed questions; lifted supported-answer rate from 78% to 92% over four controlled releases.
- Developed PyTorch text classifiers for priority routing; improved macro F1 from 0.81 to 0.89 after calibration and label audits.
- Operated production AI endpoints on Google Cloud Run with token-cost dashboards, retry policies, daily smoke tests, and alerting.
- Implemented privacy filters, source-citation audits, and negative tests to reduce unsupported replies in the customer workflow.
- Mentored four engineers in reproducible experiments, model cards, error analysis, and automated model quality gates.

Machine Learning Engineer | Northstar Digital (fictional) | **Jan 2021 - Dec 2023**

- Developed TensorFlow image and text models, including training pipelines, augmentation policies, and evaluation notebooks for three customer applications.
- Integrated NLP intent detection, entity extraction, and semantic retrieval into Python services consumed by Node.js dashboards.
- Created time-series demand forecasts with seasonal cross-validation and documented MAE/MAPE against naive baselines.
- Maintained ML feature pipelines in BigQuery and deployed repeatable containers through CI/CD for staging and production.
- Designed a lightweight medical-image triage research pilot, explicitly evaluated on de-identified public data rather than clinical use.

ENGINEERING DEPTH AND IMPACT

TECHNICAL DELIVERY DETAILS

- Training and data: stratified splits, leakage checks, annotated hold-outs, class-weight sensitivity, reproducible experiments and immutable evaluation datasets.
- LLM engineering: LangChain retrievers, LangGraph conditional flow, OpenRouter-compatible chat API clients, structured outputs, tool error handling and budget-aware batching.
- Quality methodology: source-level citation checks, contradiction cases, model-judge disagreement samples and reviewer sign-off for high-impact changes.
- Operational discipline: audit traces with model/prompt identifiers, error-rate dashboards, provider failover and rollback readiness.

DETAILED CONTRIBUTIONS & COLLABORATION

- Led root-cause review for inaccurate assistant responses by separating retrieval misses from prompt interpretation mistakes and unsupported claims.
- Published model evaluation reports to engineering and operations leads, including per-class errors and changes in recall after dataset updates.
- Coordinated production hardening of Python services, document access authorization, retention policies and role-based review workflows.
- Used Firebase and Node.js to implement non-AI workflow interfaces consuming existing FastAPI services.
- Prepared reproducible migration plans between embeddings, with explicit recall comparisons before switching vector dimensions.

ADDITIONAL WORK SAMPLES

- Insurance-document NLP pilot: transformer intent labels and human correction loop; report included class counts, macro F1 and difficult false positives.
- Demand forecast analysis: two years of retail series, seasonality baselines, future-month cutoffs, MAE and decision-impact review.
- Public medical-image segmentation research: held-out cohorts, documented dataset origin, and no claim of hospital deployment.

SELECTED PROJECT OUTCOMES

Production knowledge assistant: LangChain + LangGraph + hosted LLMs; retrieved and cited policy passages; 38% fewer manual document lookups during a controlled six-week pilot.

Multilingual ticket routing: PyTorch + transformers; 14 classes, error review, robust macro-F1 reporting, and monitored daily batches.

Visual inspection research prototype: TensorFlow transfer learning; augmentation and patient-independent split for medical imagery; research-only experiment.

EDUCATION, METHODS AND ADDITIONAL BACKGROUND

- BSc Computer Science | National Institute of Computing (fictional), 2020
- Internal model-quality reviewer; 2024-2026: maintained evaluation runbooks, prompt regression labels, and change-control checklists
- Practical deployment: Cloud Run, Vertex AI, Docker, FastAPI, PostgreSQL, Firebase authentication
- Professional interests: interpretable AI, safe LLM applications, and operational quality engineering